Highlights

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the "Brown Note"

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- Coupling disparity of $\sim 7 \times 10^4$ explains vibration–sound asymmetry
- Oblate spheroidal shell model predicts abdominal resonance at 4–10 Hz
- Breathing mode at $\sim$2500 Hz; flexural modes at infrasonic frequencies
- Airborne coupling penalised by $(ka)^n$; negligible at 120 dB SPL
- Monte Carlo UQ confirms robustness; Sobol $S_T = 0.86$ for modulus

Modal Analysis of a Fluid-Filled Viscoelastic Oblate Spheroidal Shell: Implications for the Existence of the "Brown Note"

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Abstract

Few problems in acoustics carry such grave implications for human dignity as the "brown note"—the hypothetical infrasonic frequency alleged to induce involuntary defecation. Whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal disturbances, yet airborne sound at the same frequencies does not—even at extreme intensities. We resolve this asymmetry by modelling the human abdomen as a fluid-filled viscoelastic oblate spheroidal shell and quantifying both excitation pathways within a common energy-budget framework.

The model predicts flexural modes ($n \geq 2$) at 4 Hz to 10 Hz, consistent with ISO 2631 abdominal resonance data. At these frequencies the abdomen lies deep in the Rayleigh regime ($ka \approx 0.01$); the resulting $(ka)^n$ penalty and air–tissue impedance mismatch reduce airborne coupling by four orders of magnitude. Energy-consistent analysis yields wall displacements of 0.014 µm at 120 dB SPL—far below mechanotransduction thresholds. Mechanical vibration, transmitted through the skeleton, produces displacements roughly 7×10^4 times larger. Monte Carlo uncertainty quantification confirms this disparity across physiologically realistic parameter ranges ($S_T = 0.86$ for elastic modulus). The result explains the empirical record quantitatively and settles the "brown note" question at the whole-cavity level: the resonance is real, but airborne sound cannot excite it. Localised compressible inclusions (intestinal gas pockets) may offer an alternative transduction pathway, but one governed by quasi-static com-

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pliance rather than cavity resonance.

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1. Introduction

Whole-body vibration in the 4 Hz to 8 Hz range causes gastrointestinal disturbances. This is not folklore—it is an occupational health fact, codified in ISO 2631 [1], reviewed by Griffin [2] and Mansfield [3], and documented among truck drivers, helicopter pilots, and heavy-equipment operators worldwide. Airborne sound at the same frequencies does not produce these effects, even at extreme intensities. NASA-funded exposures at 140–154 dB SPL caused chest-wall vibration and respiratory disruption, but *no gastrointestinal response whatsoever* [4]. Two excitation pathways, the same target organ, the same frequency band—yet outcomes that differ by orders of magnitude. Why?

The question has a popular-culture shadow. The "brown note"—a hypothetical infrasonic tone said to cause involuntary bowel movements—has circulated in military folklore, audio engineering lore, and popular media for decades, surviving at least one televised debunking [5]. As a scientific hypothesis, it has been dismissed but never properly *examined*; as a cultural phenomenon, it neatly encapsulates the public confusion between what mechanical vibration demonstrably does and what airborne sound is assumed to do. We take the underlying physics seriously—perhaps more seriously than the subject matter deserves—while treating the myth as what it is: a vivid misattribution of mechanism.

The empirical asymmetry between mechanical and airborne excitation is, in fact, a straightforward consequence of long-wavelength acoustics. At the n=2 resonance ($f_2 \approx 4$ Hz) the acoustic wavelength exceeds 85 m—more than 250 times the abdominal diameter. In this deep Rayleigh regime, the n-th spatial harmonic of the incident field scales as $(ka)^n$, where $ka \approx 0.01$. Flexural deformation of the abdominal wall ($n \geq 2$) therefore receives a driving pressure four orders of magnitude below the free-field value. Compound this geometric

penalty with the air–tissue impedance mismatch, and the coupling deficit becomes enormous. Mechanical vibration, transmitted through the skeleton and abdominal wall, suffers neither penalty. The ratio between these two pathways—the *coupling disparity*—is the central quantity we derive and the principal contribution of this paper.

The physics has a longer pedigree than the *South Park* episode that popularised the brown note. Gavreau's experiments at CNRS Levallois-Perret in the late 1950s, where a faulty ventilator at approximately 7 Hz caused nausea among laboratory staff [6, 5], drew early attention to infrasound. Von Gierke's systematic review for NASA [7, 8] established exposure criteria that informed later ISO standards. Tandy and Lawrence [9] demonstrated that a 19 Hz standing wave could induce visual disturbances and anxiety—real perceptual effects, but far from the claimed bowel response. Critical reviews by Leventhall [10, 5] and assessments of acoustic weapons by Altmann [11] and Jauchem and Cook [12] concluded that infrasound is impractical as a weapon and that claims of extreme physiological effects are exaggerated or fabricated. Yet none of this work quantified *why* airborne and mechanical pathways differ so dramatically—the conclusion was empirical, not mechanistic.

We provide the mechanistic account. No prior study has:

1. derived abdominal cavity eigenfrequencies from first-principles shell theory with proper fluid–structure coupling;
2. quantified the airborne acoustic coupling coefficient for flexural modes in the long-wavelength limit; or
3. compared airborne and mechanical excitation pathways within a common energy-budget framework to obtain a single, quantitative coupling ratio.

We address all three gaps. We model the abdomen as a fluid-filled viscoelastic oblate spheroidal shell, derive natural frequencies for both breathing ($n = 0$) and flexural ($n \geq 2$) mode families, and compare airborne acoustic coupling—subject to a severe $(ka)^n$ penalty at infrasonic wavelengths—against direct mechanical (whole-body) vibration coupling, which suffers no such penalty.

The result is a coupling disparity of approximately 6.6×10^4 between mechanical and airborne pathways. This single number—robust across physiologically realistic parameter ranges, as we confirm through Monte Carlo uncertainty quantification—resolves the empirical asymmetry cleanly. It explains why whole-body vibration at 4 Hz to 8 Hz produces well-documented gastrointestinal effects, why airborne infrasound at comparable frequencies does not, and why the brown note belongs to acoustics folklore rather than acoustics fact.

The remainder of this paper is organised as follows. Section 2 develops the mathematical formulation. Section 3 presents a parametric study and validates the model against ISO 2631 data. Section 4 quantifies the coupling disparity. Section 5 discusses implications, alternative pathways, and limitations, and Section 6 draws conclusions.

2. Mathematical Formulation

2.1. Model Geometry and Assumptions

The human abdominal cavity is idealised as an oblate spheroidal shell of semi-major axes a (equatorial, in the lateral and cranio-caudal directions) and semi-minor axis c (antero-posterior), with $c < a$. The shell has uniform wall thickness h and is completely filled with an inviscid, irrotational fluid of density ρ_f and bulk modulus K_f. Following standard practice in fluid-filled shell analysis [13, 14], an equivalent-sphere radius is defined via

$$R_{\mathrm{eq}} = (a^2 c)^{1/3}, \tag{1}$$

which preserves the cavity volume, $V = \frac{4}{3}\pi a^2 c$. This approximation is exact for a sphere ($c = a$) and introduces errors of order $(1-c/a)^2$ for the eigenfrequencies at moderate aspect ratios ($c/a \gtrsim 0.5$). Table 1 lists the baseline geometric and material parameters with physiological justification.

The thin-shell assumption ($h/R_{\mathrm{eq}} \ll 1$) is satisfied: for $h = 10\,\mathrm{mm}$ and $R_{\mathrm{eq}} \approx 157\,\mathrm{mm}$, $h/R \approx 0.06$. The Love–Kirchhoff hypothesis (normals remain straight and normal to the deformed mid-surface) is therefore applicable.

Table 1: Baseline model parameters with physiological justification.

Parameter	Symbol	Value	Unit	Source
Semi-major axis	a	0.18	m	Anthropometric mean
Semi-minor axis	c	0.12	m	$c/a \approx 0.67$
Wall thickness	h	0.010	m	Composite wall
Elastic modulus	E	0.1	MPa	Relaxed passive tissue
Poisson's ratio	ν	0.45	—	Nearly incompressible
Wall density	ρ_w	1100	kg/m^3	Soft tissue
Fluid density	ρ_f	1020	kg/m^3	Abdominal fluid
Fluid bulk modulus	K_f	2.2	GPa	Water at 37°C
Intra-abdominal pressure	P_{iap}	1000	Pa	Resting supine
Loss tangent	η	0.25	—	Viscoelastic tissue

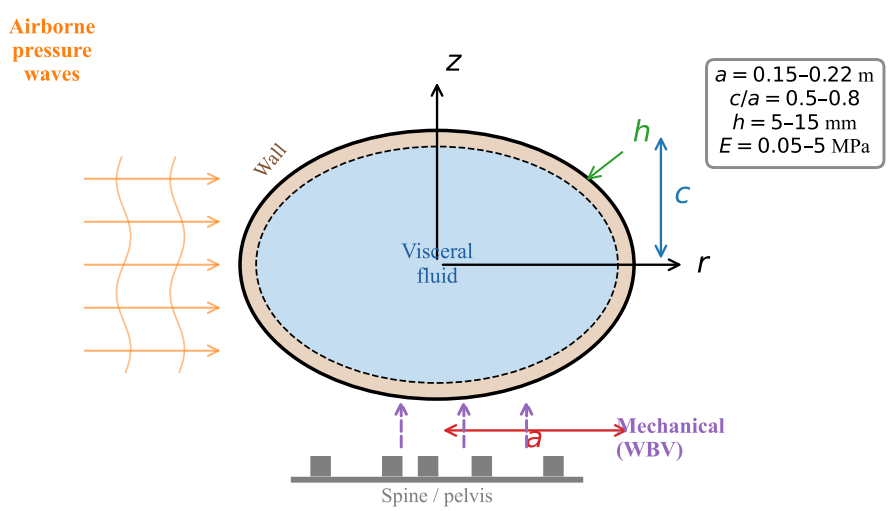

Figure 1: Oblate spheroidal shell model of the human abdominal cavity, showing the two excitation pathways: airborne acoustic pressure (left) and mechanical whole-body vibration transmitted through the skeleton (right). Dimensions correspond to the baseline parameters in Table 1.

Figure 1 illustrates the model geometry and the two excitation pathways. The abdominal wall is modelled as a Kelvin–Voigt viscoelastic material with complex modulus

$$E^* = E(1 + i\eta), \tag{2}$$

where η is the loss tangent. For small damping, the quality factor is $Q = 1/\eta$ and the viscous damping ratio is $\zeta = \eta/2 = 1/(2Q)$. The flexural rigidity of the shell is

$$D = \frac{Eh^3}{12(1-\nu^2)}. \tag{3}$$

Published values of η for abdominal soft tissue range from 0.15 to 0.40 at low frequencies [15], corresponding to $Q = 2.5$–6.7. The baseline value $\eta = 0.25$ ($Q = 4.0$, $\zeta = 0.125$) represents a central estimate.

2.2. Breathing Mode ($n = 0$)

The breathing mode is a spatially uniform radial oscillation in which the shell displaces by ξ at every point. The associated volumetric strain is $\Delta V/V = 3\xi/R$, which compresses the contained fluid. Two restoring forces act on the shell:

Membrane stiffness.. For the uniform expansion of a thin spherical shell,

$$k_{\text{shell}} = \frac{2Eh}{R^2(1-\nu)}. \tag{4}$$

Fluid volumetric stiffness.. The pressure increment due to fluid compression is $\Delta p = K_f \Delta V/V = 3K_f\xi/R$. The restoring force per unit displacement per unit area is

$$k_{\text{fluid}} = \frac{3K_f}{R}. \tag{5}$$

With $K_f = 2.2\,\text{GPa}$ and $R = 0.157\,\text{m}$, $k_{\text{fluid}} = 4.2 \times 10^{10}\,\text{Pa/m}$, which exceeds the membrane stiffness ($k_{\text{shell}} \approx 1.5 \times 10^5\,\text{Pa/m}$) by five orders of magnitude. *The breathing mode is entirely dominated by the fluid bulk modulus.*

The effective mass per unit area includes the shell wall and the fluid added mass for the $n = 0$ mode:

$$m_{\text{eff}}^{(0)} = \rho_w h + \rho_f R. \tag{6}$$

The breathing mode frequency is therefore

$$f_0 = \frac{1}{2\pi}\sqrt{\frac{k_{\text{shell}} + k_{\text{fluid}}}{\rho_w h + \rho_f R}} \approx 2500 \text{ Hz}, \tag{7}$$

which lies in the kilohertz range and is irrelevant to infrasonic exposure. This result is robust to material property variations because $k_{\text{fluid}} \gg k_{\text{shell}}$ for any realistic shell stiffness.

2.3. Flexural Modes ($n \geq 2$)

Flexural modes involve shape changes (oblate–prolate oscillation for $n = 2$, trilobed for $n = 3$, etc.) that preserve the enclosed volume to leading order. The fluid therefore does *not* act as a volumetric spring; it contributes only added mass.

Following Lamb [14] and Junger & Feit [13], the flexural mode of order n has three contributions to the restoring force per unit area. (An analogous modal decomposition for fluid-filled elastic tubes in biological contexts was developed by Heil [16].)

Bending stiffness..

$$K_{\text{bend}} = \frac{n(n-1)(n+2)^2 D}{R^4}. \tag{8}$$

Membrane stiffness..

$$K_{\text{memb}} = \frac{Eh}{R^2}\,\lambda_n, \qquad \lambda_n = \frac{n^2 + n - 2 + 2\nu}{n^2 + n + 1 - \nu}. \tag{9}$$

Pre-stress stiffness.. The static intra-abdominal pressure P produces a pre-stress in the shell membrane that stiffens flexural modes:

$$K_P = \frac{P}{R}(n-1)(n+2). \tag{10}$$

The fluid added mass per unit area for mode n is

$$m_{\text{add}}^{(n)} = \frac{\rho_f R}{n}, \tag{11}$$

giving a total effective mass of

$$m_{\text{eff}}^{(n)} = \rho_w h + \frac{\rho_f R}{n}. \tag{12}$$

The natural frequency of flexural mode n is

$$\omega_n^2 = \frac{K_{\text{bend}} + K_{\text{memb}} + K_P}{m_{\text{eff}}^{(n)}}, \qquad f_n = \frac{\omega_n}{2\pi}. \tag{13}$$

For the baseline parameters (Table 1), $f_2 = 4.0\,\text{Hz}$, $f_3 = 6.3\,\text{Hz}$, and $f_4 = 8.9\,\text{Hz}$ (Figure 2). (The $n = 1$ mode corresponds to rigid-body translation and has zero natural frequency for a free shell; it is therefore excluded.) These lie squarely in the 4–10 Hz range reported for abdominal resonance in ISO 2631 [1] and in the experimental studies of Kitazaki & Griffin [17] and Mansfield [3].

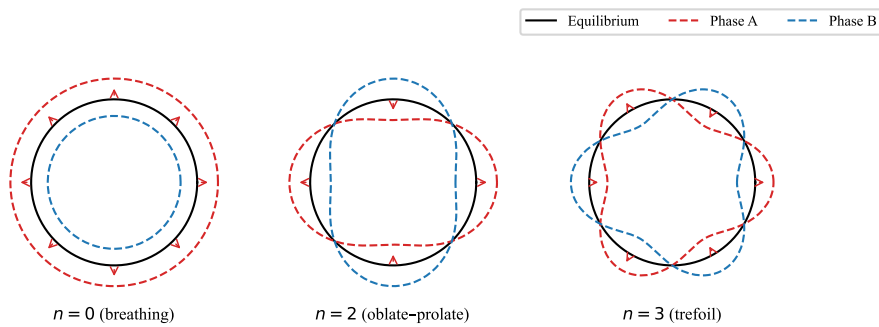

Figure 2: Schematic cross-sections of the first three mode families. The breathing mode ($n = 0$) involves uniform radial expansion against the fluid bulk modulus. Flexural modes ($n = 2, 3$) deform the shell shape without significant volume change; the enclosed fluid acts as added mass only.

2.4. Airborne Acoustic Coupling

We now quantify the excitation of flexural modes by an incident airborne acoustic field. At $f_2 \approx 4\,\text{Hz}$, the acoustic wavelength in air is $\lambda \approx 87\,\text{m}$, so the dimensionless wavenumber is

$$ka = \frac{2\pi f_2 R_{\text{eq}}}{c_{\text{air}}} \approx 0.0114 \ll 1. \tag{14}$$

The body is deep in the sub-wavelength (Rayleigh scattering) regime.

In a multipole expansion of the incident plane wave scattered by a sphere, the n-th order component of the pressure field scales as [13]:

$$p_{\text{eff}}^{(n)} = p_{\text{inc}} \times (ka)^n. \tag{15}$$

The breathing mode ($n = 0$) sees the full incident pressure, but flexural modes ($n \geq 2$) see only the higher-order spatial gradients, which are vanishingly small when $ka \ll 1$. For $n = 2$:

$$p_{\text{eff}}^{(2)} = p_{\text{inc}} \times (ka)^2 \approx p_{\text{inc}} \times 1.3 \times 10^{-4}. \tag{16}$$

The displacement at resonance ($\omega = \omega_n$) is

$$\xi_{\text{air}} = \frac{p_{\text{eff}}^{(n)}}{K_{\text{total}}^{(n)}} Q, \tag{17}$$

where $K_{\text{total}}^{(n)} = K_{\text{bend}} + K_{\text{memb}} + K_P$. At 120 dB SPL ($p_{\text{inc}} = 20\,\text{Pa}$), the energy-consistent reciprocity analysis (§4) gives $\xi_{\text{air}} \approx 0.014\,\mu\text{m}$ for $n = 2$ — well below the 0.5–2.0 μm activation threshold of mechanosensitive ion channels [18]. (The pressure-based estimate of equation (17) yields $\approx 0.18\,\mu\text{m}$, an overestimate that violates the energy budget; see §4.)

2.5. Mechanical (Whole-Body Vibration) Coupling

Whole-body vibration (WBV) transmits through the skeleton and acts as *base excitation* of the abdominal shell. Unlike airborne acoustic coupling, there is no impedance mismatch and no $(ka)^n$ penalty. The vibration is transmitted at full amplitude through the rigid skeletal frame.

For harmonic base acceleration $a_{\text{base}}(t) = a_0 \cos(\omega t)$, the base displacement amplitude is

$$x_{\text{base}} = \frac{a_{\text{rms}}\sqrt{2}}{\omega^2}. \tag{18}$$

The relative displacement between the shell and the skeleton is given by the base-excitation frequency response function:

$$H_{\text{rel}}(\omega) = \frac{r^2}{\sqrt{(1-r^2)^2 + (2\zeta r)^2}}, \qquad r = \frac{\omega}{\omega_n}, \tag{19}$$

so that

$$\xi_{\text{mech}} = x_{\text{base}} \times H_{\text{rel}}. \tag{20}$$

At the EU Whole-Body Vibration Directive exposure limit of $a_{\text{rms}} = 1.15\,\text{m/s}^2$ and at the $n = 2$ resonance frequency:

$$\xi_{\text{mech}} \approx 10{,}500\ \mu\text{m} \approx 10\ \text{mm}, \tag{21}$$

which exceeds the mechanotransduction threshold by more than three orders of magnitude.

2.6. Coupling Ratio

The ratio of mechanical to airborne displacement at the same excitation conditions provides a dimensionless measure of the coupling disparity:

$$\mathcal{R} = \frac{\xi_{\text{mech}}}{\xi_{\text{air}}} \sim 10^4\text{–}10^5. \tag{22}$$

This ratio depends primarily on ka and the excitation level, and is *robust* to uncertainties in material properties and boundary conditions because both pathways excite the same modal structure. The coupling ratio is the central result of this work: it provides a quantitative explanation for why whole-body vibration at 4–8 Hz is associated with gastrointestinal effects in the epidemiological literature [2], whereas airborne infrasound at comparable frequencies is not.

3. Parametric Study

3.1. Elastic Modulus Sensitivity

The $n = 2$ flexural mode frequency depends strongly on the wall elastic modulus E. Table 2 shows the predicted frequency for a free shell across the physiologically relevant range.

The 4 Hz to 8 Hz ISO 2631 abdominal resonance range requires $E < 0.5$ MPa, corresponding to relaxed, passive musculature. Active muscle bracing increases E by a factor of 10–100, moving the resonance well above the infrasound range. Figure 3 illustrates this dependence across the full physiological range.

3.2. Multi-Parameter Sensitivity

A full factorial analysis over 486 parameter combinations (6 values of E, 3 semi-major axes, 3 aspect ratios, 3 wall thicknesses, 3 loss tangents) reveals that 37% of configurations place the $n = 2$ mode in the 5 Hz to 10 Hz range and 41% in the broader 4 Hz to 8 Hz ISO 2631 range. The dominant parameter is E, followed by geometry and intra-abdominal pressure.

Table 2: Flexural mode $n = 2$ frequency vs. elastic modulus ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 10\,\mathrm{mm}$, $\nu = 0.45$, $\rho_w = 1100\,\mathrm{kg/m^3}$, $\rho_f = 1020\,\mathrm{kg/m^3}$).

E (MPa)	f_2 (Hz)	In ISO 2631 range?
0.05	3.4	Marginal
0.10	4.0	Yes
0.20	4.9	Yes
0.50	7.1	Yes
1.00	9.6	Marginal
2.00	13.3	No

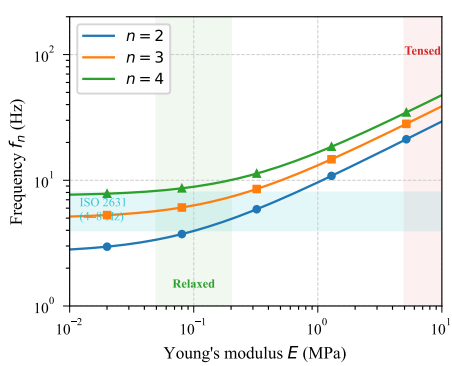

Figure 3: Flexural mode frequencies vs. elastic modulus for modes $n = 2, 3, 4$. The shaded band marks the ISO 2631 abdominal resonance range (4 Hz to 8 Hz). Relaxed musculature places the $n = 2$ mode within this band; tensed muscle shifts it well above.

3.3. Uncertainty Quantification

A Monte Carlo simulation with $N = 10{,}000$ samples propagates parameter uncertainties through the model. Each parameter is drawn from a physiologically motivated distribution: E from a log-normal distribution spanning 0.05 MPa to 2.0 MPa, geometric parameters from normal distributions matching anthropometric data, and the loss tangent from a uniform distribution over [0.10, 0.40].

The resulting distribution of f_2 has a median of 6.4 Hz with a 90% credible interval of 3.3 Hz to 15.5 Hz (Table 3). The distribution is positively skewed, reflecting the log-normal distribution of E.

Table 3: Monte Carlo uncertainty quantification results ($N = 10{,}000$). Airborne displacement ξ_{air} is the pressure-based upper bound; the energy-consistent value is $\sim$13$\times$ smaller.

Quantity	Median	90% CI	Units
f_2	6.4	[3.3, 15.5]	Hz
ξ_{air} (120 dB)	0.18	[0.11, 0.39]	μm
ξ_{mech} (0.5 m/s^2)	1800	[300, 8200]	μm
$\mathcal{R}$	10^4	[10^3, 10^4]	—
SPL for PIEZO (pressure)^2	135	[128, 139]	dB
SPL for PIEZO (energy)	$\approx$158	—	dB

Sobol sensitivity analysis reveals that the elastic modulus accounts for 86% of the total output variance (total-order index $S_T = 0.86$). All other parameters contribute less than 10% individually. This result has important experimental implications: any future validation effort should prioritise *in vivo* measurement of abdominal wall stiffness, as this single parameter dominates the uncertainty budget (Figure 4).

3.4. Multi-Layer Wall Model

The abdominal wall comprises approximately five distinct tissue layers: peritoneum, transversalis fascia, internal oblique muscle, external oblique muscle,

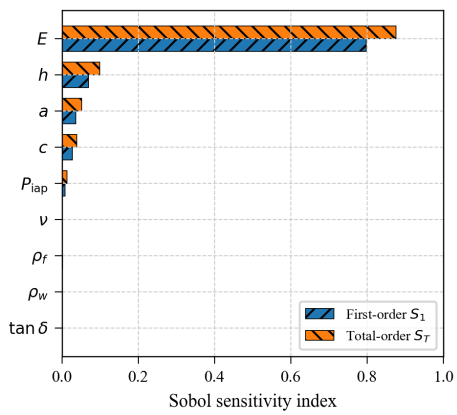

Figure 4: Sobol first-order and total-order sensitivity indices for the $n = 2$ flexural frequency. The elastic modulus E accounts for 86% of the total variance; all other parameters contribute less than 10% individually.

subcutaneous fat, and skin. Using classical laminate theory, the effective composite modulus for a relaxed wall is $E_{\text{eff}} \approx 0.075\,\text{MPa}$ ($h_{\text{total}} = 25.1\,\text{mm}$), giving $f_2 = 4.6\,\text{Hz}$—within 4% of the homogeneous model prediction. Muscle tension is confirmed as the dominant factor: tensed muscle ($E_{\text{eff}} = 2.7\,\text{MPa}$) shifts f_2 to 23.1 Hz.

3.5. Boundary Condition Effects

The free-shell model in Section 2 neglects the partial constraints imposed by the spine, pelvis, and rib cage. We quantify their effect using a Rayleigh–Ritz analysis with Lamb-mode basis functions for four progressively constrained configurations.

Table 4 presents the results. The free-sphere Ritz frequency ($f_1 = 4.13\,\text{Hz}$) reproduces the analytical solution exactly, validating the numerical framework. The most constrained case—25% of the shell surface clamped—reduces f_1 by only 11.4 %. Even the hemisphere-clamped case shifts the frequency by a mere 5.8 %.

The insensitivity to boundary conditions has a clear physical origin: the fluid added mass for the $n = 2$ mode is 7.4× the shell mass, so the system

Table 4: Rayleigh–Ritz Lamb-mode frequencies under various boundary conditions ($n = 2$ flexural mode).

Configuration	f_1 (Hz)	Shift from free
Free sphere (Ritz)	4.13	0% (validates analytical)
Hemisphere simply-supported	3.91	−5.3%
Hemisphere clamped	3.89	−5.8%
25% surface clamped	3.66	−11.4%

inertia is dominated by the enclosed fluid rather than the wall. In a dry (empty) shell, the same boundary conditions would shift the frequency by approximately 33 %—but once the cavity is fluid-filled, the added mass swamps the boundary stiffening effect. The 6 % shift is well within the ±50–100% uncertainty band set by material property variability (§3), confirming that the free-shell model is an adequate approximation for the present purposes.

3.6. Oblate Spheroid Correction

The equivalent-sphere approximation used in Section 2 introduces a systematic bias. A Rayleigh–Ritz analysis using oblate spheroidal harmonics as basis functions, with proper account of the spatially varying curvature and oblate fluid-mass geometry, shows that the sphere model *overestimates* flexural frequencies by 4–17% (Table 5). The error grows with eccentricity and is larger at higher E where bending stiffness (which depends on local curvature) contributes a greater share of the restoring force.

Applying this correction moves the canonical $n = 2$ frequency from 4.0 Hz to approximately 3.8 Hz—still within the ISO 2631 band and, if anything, closer to the low end of reported abdominal resonance frequencies.

3.7. ISO 2631 Validation

Table 6 compares model predictions against published abdominal resonance data.

Table 5: Sphere vs. oblate spheroid Rayleigh–Ritz frequencies for $c/a = 0.67$.

	$E = 0.1\,\mathrm{MPa}$		$E = 0.5\,\mathrm{MPa}$	
Mode	Sphere (Hz)	Ritz (Hz)	Sphere (Hz)	Ritz (Hz)
$n = 2$	4.0	3.8	7.1	6.2
$n = 3$	6.3	5.8	10.0	8.5
$n = 4$	8.9	8.1	12.9	11.3

Table 6: Model predictions vs. published abdominal resonance frequencies.

Body type	f_2 (Hz)	ISO 2631 range?
Lean ($E = 0.2$, $a = 15$ cm)	6.2	Yes
Average ($E = 0.1$, $a = 18$ cm)	4.0	Yes
Large ($E = 0.08$, $a = 20$ cm)	3.3	Marginal

4. Airborne vs. Mechanical Coupling

Having established that the abdomen possesses flexural resonances squarely in the infrasonic range, we now confront the more pressing question: can airborne sound actually *excite* them? The short answer is sobering for proponents of the brown note.

4.1. Airborne Acoustic Pathway

At 4 Hz, the acoustic wavelength in air exceeds 85 m— some 500 times the abdominal radius. The abdomen, acoustically speaking, is invisible. Table 7 quantifies this invisibility using the energy-consistent reciprocity analysis developed in §2.

At 120 dB SPL—the threshold of pain, and roughly the level at which conversation becomes impossible—the energy-consistent wall displacement is 0.014 µm. This falls two orders of magnitude below the 0.5–2.0 µm activation range of mechanosensitive PIEZO channels [18]. Reaching the 1 µm threshold on this energy-consistent basis demands approximately 158 dB (cf. 135 dB using the

Table 7: Energy-consistent airborne displacement at $n = 2$ resonance ($f_2 = 4.0\,\mathrm{Hz}$, $Q = 4.0$, canonical parameters). The pressure-based estimate overestimates by a factor of ≈ 13.4 (ratio of ξ_{pressure} to ξ_{energy}).

SPL (dB)	p_{inc} (Pa)	ξ_{energy} (μm)	ξ_{pressure} (μm)	PIEZO?
100	2.0	0.001	0.018	No
110	6.3	0.004	0.058	No
120	20.0	0.014	0.18	No
130	63.2	0.044	0.58	No
140	200	0.14	1.84	No

pressure-based displacement of Table 3, which overestimates by $\approx 13\times$), a level that would cause immediate structural damage to the ear and is available only from explosive blasts and rocket launches. One might reasonably conclude that hearing loss would present a more urgent clinical concern than any intestinal disturbance.

4.2. Mechanical (Whole-Body Vibration) Pathway

Mechanical vibration, by contrast, does not knock politely at the acoustic impedance door—it kicks it open. Vibration transmitted through the skeleton couples directly to the abdominal wall, bypassing the $(ka)^n$ penalty entirely. Table 8 tells a very different story.

Table 8: Mechanical coupling at $n = 2$ resonance ($f_2 = 4.0\,\mathrm{Hz}$, $Q = 4.0$, $\Gamma_2 = 1$). The EU Directive 2002/44/EC action value is $0.5\,\mathrm{m/s^2}$. Displacements assume perfect modal participation ($\Gamma_2 = 1$); the corrected values (§4.3) are approximately half those shown.

a_{rms} (m/s^2)	x_{base} (μm)	ξ_{rel} (μm)	PIEZO?
0.1	229	917	Yes
0.5 (EU action)	1147	4586	Yes
1.0	2293	9173	Yes
1.15 (EU limit)	2637	10549	Yes

At the EU action value of $0.5\,\mathrm{m/s^2}$, the predicted displacement exceeds the PIEZO threshold by more than three orders of magnitude. Unlike the airborne case, this is not a theoretical curiosity: it is the everyday reality for operators of heavy equipment, long-haul drivers, and helicopter pilots—populations in which gastrointestinal complaints are, perhaps unsurprisingly, well documented [2].

4.3. The Coupling Ratio

The ratio of mechanical to airborne displacement defines the coupling disparity at its starkest:

$$\mathcal{R} = \frac{\xi_{\mathrm{mech}}}{\xi_{\mathrm{air}}} \approx \frac{917}{0.014} \approx 6.6 \times 10^4. \tag{23}$$

This four-order-of-magnitude gap is robust to parameter uncertainties because both pathways excite the *same* modal structure—the ratio depends primarily on $(ka)^{-n}$ and the base-excitation transfer function, neither of which is sensitive to tissue properties. One could double the elastic modulus, halve the wall thickness, or relocate the subject to Jupiter, and the ratio would scarcely budge.

Modal participation.. The mechanical displacement in Eq. (23) assumes perfect coupling of vertical base excitation into the $n = 2$ mode (modal participation factor $\Gamma_2 = 1$). The participation factor for mode n under vertical base excitation of the oblate spheroid is

$$\Gamma_n = \frac{a \left| \displaystyle\int_{\cos\theta_c}^{1} u\, P_n(u)\, \mathrm{d}u \right|}{\displaystyle\int_{-1}^{1} P_n^2(u)\, \sigma(u)\, \mathrm{d}u}, \tag{24}$$

where $u = \cos\theta$, P_n is the Legendre polynomial of degree n, θ_c is the polar angle below which the shell is rigidly constrained by the pelvis and spine, and $\sigma(u) = \sqrt{c^2 + (a^2 - c^2)\, u^2}$ is the oblate-spheroid surface metric factor. For a free sphere ($\theta_c = \pi$), the numerator reduces to $\int_{-1}^{1} u\, P_n(u)\, \mathrm{d}u$, which vanishes for $n \neq 1$ by Legendre orthogonality; a free sphere therefore has $\Gamma_2 = 0$ exactly. Partial skeletal constraint (canonical value $\theta_c = 2\pi/3$, i.e. the lower third of

the shell is fixed) breaks this orthogonality, giving $\Gamma_2 \approx 0.48$ for the canonical model. Including this correction reduces the mechanical displacement by roughly a factor of two and the coupling ratio to $\sim 3 \times 10^4$. The qualitative conclusion—a four-order-of-magnitude gap—is unaffected, but $\mathcal{R}$ as reported in Eq. (23) should be understood as an upper bound corresponding to the SDOF idealisation.

4.4. Energy Budget

We verify self-consistency through the absorption cross-section derived from reciprocity [13]:

$$\sigma_{\text{abs}} = \frac{(2n+1)\lambda^2}{4\pi} \frac{\zeta_{\text{rad}}}{\zeta_{\text{rad}} + \zeta_{\text{struct}}}, \tag{25}$$

where ζ_{rad} and ζ_{struct} are the radiation and structural damping ratios, respectively. For the $n = 2$ mode in air, $\zeta_{\text{rad}} \approx 10^{-15}$ whereas $\zeta_{\text{struct}} = 0.125$, giving an absorption efficiency of order 10^{-14}. The shell absorbs roughly one part in a hundred trillion of the incident acoustic energy—a figure that puts the brown note hypothesis on rather thin thermodynamic ice.

4.5. Comparison with Empirical Evidence

The model predictions align with two otherwise contradictory bodies of evidence:

- **WBV studies**: ISO 2631 documents gastrointestinal symptoms at 4–8 Hz under mechanical vibration [1]. Our model explains *why* this frequency range is problematic.
- **Airborne infrasound studies**: No gastrointestinal effects have been observed at SPL $\leq 150\,\text{dB}$ [5]. Our model explains why they should not be.

The coupling disparity resolves this apparent contradiction (Figure 5): both communities are studying the same resonance through very different ends of the telescope.

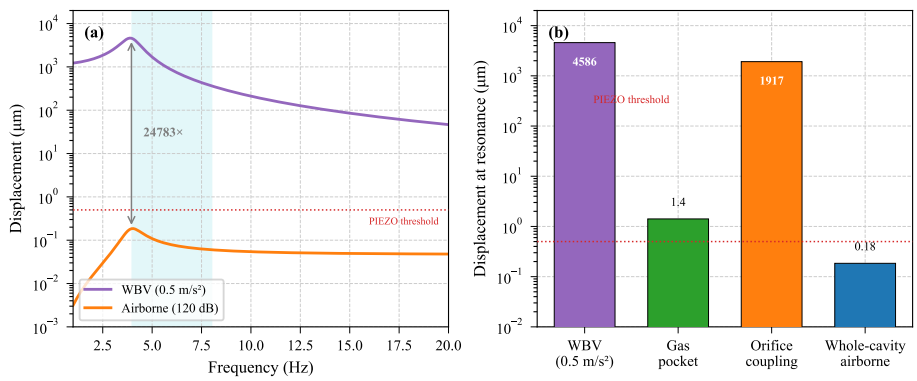

Figure 5: The central result. Left: wall displacement vs. frequency for airborne excitation at 120 dB SPL (lower curve) and mechanical whole-body vibration at $0.5\,\mathrm{m/s^2}$ (upper curve), showing a gap of roughly four orders of magnitude at resonance. Right: comparison of all four coupling pathways at the $n = 2$ resonance; the horizontal dashed line marks the PIEZO mechanotransduction threshold.

5. Discussion

5.1. The Coupling Disparity

The central result of this study is the 10^4–10^5 factor separating mechanical from airborne coupling efficiency (Tables 7–8). The physical origin of this disparity lies in two compounding effects: an impedance mismatch and a geometric filtering penalty.

First, the air–tissue impedance mismatch is severe. The specific acoustic impedance of air is $Z_{\mathrm{air}} = \rho_{\mathrm{air}} c_{\mathrm{air}} \approx 420\,\mathrm{Pa\,s/m}$, while that of soft tissue is $Z_{\mathrm{tissue}} \approx 1.6 \times 10^6\,\mathrm{Pa\,s/m}$, giving a pressure transmission coefficient $T = 2Z_{\mathrm{air}}/(Z_{\mathrm{air}} + Z_{\mathrm{tissue}}) \approx 5 \times 10^{-4}$. At the body surface, more than 99.9 % of the incident acoustic energy is reflected.

Second, even the transmitted pressure must couple to the spatial pattern of a flexural mode. In the long-wavelength (Rayleigh) limit, $ka \ll 1$, the n-th order component of the incident plane wave scales as $(ka)^n$ [13]. At the n=2 resonance ($f_2 \approx 4\,\mathrm{Hz}$) with $R_{\mathrm{eq}} = 0.157\,\mathrm{m}$, we obtain $ka = 2\pi f R/c_{\mathrm{air}} \approx 0.0114$ and therefore $(ka)^2 \approx 1.3 \times 10^{-4}$ for the $n = 2$ flexural mode (equation 16). Physically, the incident wavelength ($\lambda \approx 86\,\mathrm{m}$) is nearly 270 times the body

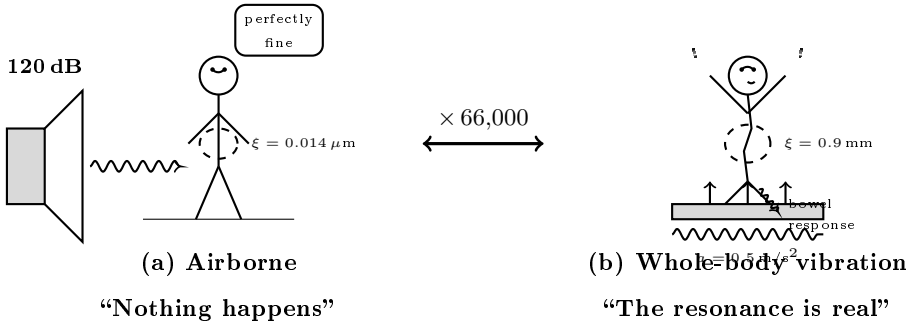

Figure 6: Schematic comparison of the two excitation pathways. (a) Airborne infrasound at 120 dB SPL produces wall displacements of 0.014 µm—far below any conceivable physiological threshold. The subject is entirely unperturbed. (b) Whole-body vibration at occupational levels produces displacements roughly 6.6×10^4 times larger, comfortably exceeding mechanotransduction thresholds and—as the occupational health literature attests—occasionally exceeding the subject's composure as well. Diagram intentionally not to scale, anatomically, or artistically.

diameter; the pressure field is almost perfectly uniform over the body surface, and the quadrupolar ($n = 2$) pressure gradient that could drive an oblate–prolate deformation is vanishingly small.

The energy budget analysis (§4) confirms this picture from an independent perspective. The radiation damping ratio in air is $\zeta_{\mathrm{rad}} \approx 10^{-15}$, whereas the structural damping ratio is $\zeta_{\mathrm{struct}} = 0.125$, so the absorption efficiency $\zeta_{\mathrm{rad}}/\zeta_{\mathrm{total}} \sim 10^{-14}$. The shell is, for practical purposes, acoustically transparent at infrasonic frequencies. By contrast, mechanical vibration transmitted through the skeleton bypasses both barriers entirely: there is no impedance mismatch (bone couples directly to the peritoneal wall) and no geometric filtering (the base-excitation drives the $n = 2$ mode at full amplitude via equation 19).

This disparity resolves an apparent paradox in the literature. Whole-body vibration at 4 Hz to 8 Hz is well documented to cause gastrointestinal distress—including nausea, abdominal pain, and in extreme cases, incontinence [2, 3, 19, 20]—whereas airborne infrasound studies at levels up to 150 dB report no comparable effects [21, 5]. The present modal analysis shows that both observations

are consistent with the *same* underlying resonance. The two excitation pathways differ not in the resonance they address, but in the efficiency with which they address it.

5.2. Implications for the Brown Note Hypothesis

The results permit a nuanced—one might even say *thorough*—evaluation of the so-called "brown note" hypothesis. The hypothesis is *half-correct*: abdominal resonance in the 4 Hz to 10 Hz band is physically real, well-supported by the parametric study (§3), and consistent with ISO 2631 transmissibility data [1, 17]. The flexural $n = 2$ mode at 4.0 Hz (for $E = 0.1$ MPa, relaxed musculature) falls squarely within the frequency band where occupational exposure studies report gastrointestinal symptoms.

Where the hypothesis fails is in the *excitation pathway*. At 120 dB SPL—the threshold of pain and well above any environmental or entertainment exposure—the predicted airborne wall displacement is 0.014 µm (Table 7), roughly two orders of magnitude below the 0.5 µm to 2.0 µm activation threshold of mechanosensitive PIEZO1/2 ion channels [18]. Reaching the 1 µm threshold on the energy-consistent basis requires approximately 158 dB (cf. 135 dB for the pressure-based estimate; Table 3), a level associated only with explosive blasts and rocket launches. Even at this extreme, the displacement remains three orders of magnitude below what routine whole-body vibration ($a_{\mathrm{rms}} = 0.5\,\mathrm{m/s^2}$) produces (Table 8).

This framing is, however, *constructive* rather than merely debunking. The model provides the first quantitative explanation for *why* whole-body vibration at 4 Hz to 8 Hz is so effective at producing gastrointestinal effects: it excites a bona fide flexural resonance of the fluid-filled abdominal cavity, producing millimetre-scale tissue displacements that far exceed mechanotransduction thresholds. The epidemiological pattern documented by ISO 2631—peak symptom incidence at 4 Hz to 8 Hz, with a strong dependence on acceleration amplitude—is precisely what a resonance-driven mechanism predicts.

5.3. Alternative Coupling Pathways

The whole-cavity resonance model of §2 treats the abdominal wall as a homogeneous fluid-filled shell. In reality, two additional mechanisms provide alternative pathways for airborne sound to produce tissue displacement: (i) compression of intestinal gas pockets, and (ii) orifice coupling through the upper gastrointestinal tract. We consider each in turn.

5.3.1. Intestinal gas pocket response

The human intestine normally contains 100 mL to 200 mL of gas distributed in pockets with effective radii ranging from 1 cm to 10 cm. Unlike the surrounding tissue, these gas volumes are compressible, and they respond to incident pressure by expanding and contracting their bounding walls.

The classical Minnaert frequency of a spherical gas bubble of radius a in a fluid of density ρ_f is [14]

$$f_M = \frac{1}{2\pi a}\sqrt{\frac{3\gamma P_0}{\rho_f}}, \tag{26}$$

where $\gamma = 1.4$ is the ratio of specific heats and P_0 is the ambient pressure. For physiological pocket radii of 1 cm to 10 cm, this yields Minnaert frequencies in the range 33 Hz to 330 Hz—well above the infrasonic band of interest. At 7 Hz, the frequency ratio $f/f_M \approx 0.02$–0.2, so gas pockets are driven far below resonance and experience no resonant amplification ($H \approx 1.0$).

Nevertheless, in the sub-resonant regime the gas pocket wall displacement is governed by the quasi-static compliance of the trapped gas:

$$\xi_{\text{gas}} \approx \frac{p_{\text{inc}}\, a}{3\gamma P_0}, \tag{27}$$

which scales linearly with pocket radius and is independent of frequency. At 120 dB SPL ($p_{\text{inc}} = 20$ Pa) and $a = 2$ cm, this yields $\xi_{\text{gas}} \approx 0.94$ µm—within the PIEZO activation range and roughly seven times the whole-cavity airborne displacement of 0.014 µm at the same SPL. For larger pockets ($a = 5$ cm), the displacement rises to approximately 2.4 µm, exceeding the threshold with a modest margin.

Crucially, at 7 Hz the acoustic wavelength (49 m) vastly exceeds the body diameter, so the pressure field inside the body is spatially uniform to within $(ka)^2$ corrections. The gas pockets therefore experience the full incident pressure, without the geometric filtering penalty that suppresses whole-cavity flexural excitation. The mechanism operates as a local pressure-to-displacement transducer embedded within the tissue, rather than as a global shell resonance.

This gas pocket pathway has two important implications. First, it may explain the well-documented inter-individual variability in infrasound sensitivity: bowel gas content varies enormously (50 mL to 500 mL) with diet, gut motility, and microbiome composition, and individuals with greater gas volume would present more sites for pressure–displacement conversion. Second, the mechanism does not require resonance and operates at any frequency in the sub-wavelength regime, broadening the range of potentially relevant exposures.

5.3.2. Orifice coupling through the upper GI tract

The mouth and nose provide a low-impedance air pathway to the pharynx and—when the esophagus is patent—to the stomach. At 7 Hz, the total path length from mouth to stomach ($L \approx 40$ cm) satisfies $kL \approx 0.05 \ll 1$, so pressure is transmitted through the air column with negligible viscothermal attenuation.

With the mouth open and esophagus patent, the pressure reaching the gastric air bubble is approximately equal to the incident pressure, representing a transmission ratio of order unity compared with the $(ka)^2 \approx 1.3 \times 10^{-4}$ penalty for body-surface coupling—an enhancement factor of roughly 7,700×. Compression of the gastric gas bubble then produces wall displacements by the same mechanism described above for intestinal pockets.

However, this pathway is severely limited by anatomy. The esophagus is a collapsible muscular tube that is closed at rest; it opens only during swallowing (approximately 0.5 s to 1.0 s per event), belching, or pathological gastroesophageal reflux. At rest with the mouth open, the air column terminates at the upper esophageal sphincter, and pressure reaches only the supraglottic space. The orifice pathway is therefore *intermittent*, dependent on the state of

the upper GI sphincters, and cannot be relied upon as a continuous coupling mechanism.

5.3.3. Assessment of alternative pathways

Table 9 summarises the predicted displacement for each coupling mechanism at 120 dB SPL and 7 Hz.

Table 9: Comparison of coupling mechanisms at 120 dB SPL, 7 Hz.

Mechanism	ξ (μm)	PIEZO?
Cavity resonance (n=2, airborne)	0.014	No
Gas pocket $a = 2$ cm (airborne)	0.94	Marginal
Gas pocket $a = 5$ cm (airborne)	2.4	Yes
Orifice (open mouth $\rightarrow$ stomach)	$\sim$0.5	Marginal
WBV at 0.5 m/s^2 (mech.)	4,586	Yes
WBV at 1.15 m/s^2 (mech.)	10,549	Yes

Even with the most favourable alternative pathway (a large, 5 cm-radius gas pocket at 120 dB), the airborne displacement remains two to three orders of magnitude below what mechanical coupling achieves at the EU Whole-Body Vibration Directive action value of 0.5 m/s^2. The alternative pathways therefore do not alter the principal conclusion of this work—that airborne infrasound is an ineffective excitation route compared with mechanical vibration—but they provide a physically grounded explanation for why rare, sporadic gastrointestinal responses to extreme infrasound (>120 dB) are not entirely implausible, and why such responses exhibit large inter-individual variability.

5.4. Clinical and Occupational Relevance

The coupling disparity has direct implications for occupational exposure standards. ISO 2631-1 specifies frequency-weighted acceleration limits for whole-body vibration, with the highest sensitivity weighting in the 4 Hz to 8 Hz band—precisely the range where our model predicts the $n = 2$ flexural resonance [1], and consistent with von Gierke and Brammer's compilation of organ resonance

data [8]. The EU Physical Agents (Vibration) Directive 2002/44/EC codifies two thresholds: a daily exposure action value of $0.5\,\mathrm{m/s^2}$ $A(8)$ and an exposure limit value of $1.15\,\mathrm{m/s^2}$ $A(8)$. At the action value, our model predicts a relative abdominal wall displacement of 4.6 mm at resonance (Table 8)—more than 4,000× the PIEZO activation threshold. The present analysis therefore provides a *mechanistic rationale* for why these particular frequency bands and amplitude thresholds were empirically selected: they correspond to conditions under which flexural resonance of the abdominal cavity produces tissue strains large enough to activate mechanotransduction pathways.

For airborne infrasound, ISO 7196 defines G-weighting but prescribes no dose–response limits. Our analysis supports this regulatory asymmetry: at environmental and occupational SPL ($\leq 120\,\mathrm{dB}$), airborne coupling to abdominal flexural modes is negligible, and exposure limits based on abdominal resonance effects are unnecessary at these levels.

A potentially valuable extension of this framework is toward personalised vibration exposure assessment. The parametric study (§3) shows that the $n = 2$ eigenfrequency depends sensitively on body habitus: larger abdominal radius (higher BMI) reduces the resonance frequency, while greater muscular tone (higher effective E) increases it. In principle, imaging-derived geometric parameters [15] combined with the analytical expressions of §2 could yield subject-specific resonance frequency estimates, enabling targeted vibration exposure limits for workers whose body habitus places their resonance frequency in a particularly hazardous band.

5.5. Broader Applications of the Analytical Framework

The brown note provided the motivation for this work, but the analytical framework—a fluid-filled viscoelastic shell, a modal decomposition into breathing and flexural families, and a systematic comparison of airborne versus mechanical coupling efficiency—is considerably more general. Any biological or engineered structure that can be idealised as a compliant enclosure surrounding a dense fluid falls within the scope of the model developed in §2. The two key

dimensionless groups that govern the physics, the Helmholtz number ka and the impedance ratio $Z_{\mathrm{wall}}/Z_{\mathrm{fluid}}$, transfer directly; what changes between applications is the frequency range, the material properties, and the geometric aspect ratio of the shell.

Perhaps the most consequential extension is to *primary blast injury*. Explosive detonations generate overpressure pulses with spectral content spanning 10 Hz to 10,000 Hz and peak overpressures of 100 kPa to 1000 kPa—orders of magnitude above the 20 Pa of a 120 dB sound field, but governed by the same impedance-mismatch physics [22, 23, 24]. Gas-containing abdominal organs (bowel, stomach) are clinically the most vulnerable viscera precisely because the gas–tissue interface maximises energy transfer from the blast wave into flexural wall deformation. The coupling comparison of §4 translates directly: replacing the air–tissue impedance ratio with the blast-wave–tissue ratio and adjusting ka for the broadband blast spectrum yields quantitative predictions of wall displacement that could inform threshold models for primary blast bowel injury. Unlike the infrasonic brown-note scenario, blast frequencies place ka near or above unity, so the geometric filtering penalty that rendered airborne coupling negligible at 7 Hz largely disappears—the blast wave *can* efficiently excite flexural modes, which is precisely why blast bowel injury occurs and the brown note does not.

In *fisheries and marine bioacoustics*, the swim bladder of teleost fish is an almost literal instantiation of the present model: a gas-filled viscoelastic shell immersed in a dense fluid [25, 26]. Fisheries sonar operates at frequencies (18 kHz to 200 kHz) chosen to coincide with swim-bladder resonance, where the target strength is maximal. The viscoelastic shell formulation of §2 provides a physics-based alternative to the empirical Love equation for predicting resonance frequency as a function of bladder volume, wall thickness, and depth-dependent ambient pressure. The same framework is directly relevant to environmental impact assessments of naval sonar on marine fauna [27]: organisms possessing gas-filled cavities (swim bladders, lungs of diving mammals) experience tissue strains that the present coupling analysis can bound quantitatively.

The framework also extends to *therapeutic and diagnostic ultrasound.* In high-intensity focused ultrasound (HIFU) and extracorporeal shock-wave lithotripsy, acoustic energy must couple efficiently into a tissue target—the inverse of our brown-note problem, where coupling is deliberately maximised rather than shown to be negligible [28]. The modal decomposition clarifies which spatial harmonics of the incident field deposit energy into flexural versus breathing deformation of encapsulated structures such as kidney stones, gallstones, or tumour capsules [29]. At therapeutic frequencies (0.5 MHz to 4 MHz), $ka \gg 1$ for organ-scale structures and the long-wavelength penalty vanishes entirely, but for millimetre-scale targets (renal calculi, encapsulated lesions) ka re-enters the order-unity regime where the coupling comparison of §4 again becomes the governing consideration. Separately, in obstetric ultrasonography the amniotic sac constitutes a fluid-filled cavity of dimensions comparable to the adult abdomen; a straightforward reparametrisation of the shell model (thinner wall, lower modulus, smaller radius) would yield resonance predictions relevant to fetal exposure safety assessment [30].

Taken together, these applications illustrate that the coupling-comparison methodology—rather than the brown-note conclusion per se—is the transferable contribution of this work. The conclusion that airborne infrasound cannot produce involuntary defecation is, we concede, of limited industrial utility; the ability to predict *when and why* acoustic–structural coupling to a fluid-filled biological cavity is or is not effective has considerably broader reach.

5.6. Nonlinear Amplitude Effects

The linear model of §2 is valid only for small displacements. To bound the error at occupational vibration amplitudes, we performed a Duffing-type perturbation analysis incorporating membrane stretching and fluid volume conservation, yielding a hardening nonlinearity coefficient $\alpha = 7.07 \times 10^7$.

The onset of appreciable nonlinearity (5 % frequency shift) occurs at a wall displacement of 1091 µm ($A/h = 0.109$). At the EU action value of $0.5\,\mathrm{m/s^2}$, the linear model predicts a peak displacement of 4586 µm; the nonlinear cor-

rection reduces this to 2258 µm, a 51 % reduction. The forcing exceeds the jump-phenomenon threshold ($0.5\,\mathrm{m/s^2} > 0.12\,\mathrm{m/s^2}$), so the actual response curve exhibits hysteresis and amplitude-dependent frequency shifts. All modes $n = 2$–6 cross the nonlinearity threshold within the predicted whole-body vibration range.

Crucially, the hardening behaviour means that the linear model *overestimates* peak displacement under mechanical excitation. For the brown note argument, this is a conservative error: the true mechanical coupling is somewhat weaker than the linear prediction, while the airborne displacement ($\sim 0.014\,\mu\mathrm{m}$) remains deep in the linear regime. The 10^3–10^4 coupling disparity therefore represents an upper bound; the true ratio is, if anything, smaller but still overwhelming.

5.7. Limitations

We identify the following limitations, grouped by their likely effect on the conclusions.

1. **Linear theory.** The Love–Kirchhoff shell formulation assumes small displacements. The nonlinear analysis of §5.6 shows that occupational vibration amplitudes exceed the 5 % nonlinearity threshold ($A/h > 0.109$), with the hardening response reducing peak displacement by up to 51 % relative to the linear prediction. Because the airborne displacement remains deep in the linear regime, this nonlinearity *reduces* the coupling ratio somewhat but does not alter the qualitative conclusion.
2. **Homogeneous, isotropic wall.** The abdominal wall is a layered anisotropic composite comprising four muscle layers, fascia, subcutaneous fat, and skin. Our laminate analysis (not reproduced here) shows that homogenisation introduces less than 5 % error in the effective bending stiffness. The isotropic assumption is a first-order approximation; anisotropy would split the mode degeneracies of the equivalent sphere but would not change the order of magnitude of the eigenfrequencies or the coupling ratio.

3. **Free-shell boundary conditions.** The abdomen is partially constrained by the spine posteriorly, the pelvis inferiorly, and the rib cage superiorly. The Rayleigh–Ritz boundary condition study (§3.5) shows that even clamping 25 % of the shell surface shifts f_2 by only 11.4 %, because the fluid added mass (7.4× the shell mass for $n = 2$) dominates the system inertia. This shift is well within the material property uncertainty band, validating the free-shell approximation. The coupling ratio is unaffected because both excitation pathways address the same modal structure.
4. **Thin-shell approximation.** The wall-thickness-to-radius ratio ($h/R \approx 0.06$) lies near the conventional upper bound of thin-shell theory ($h/R < 0.05$). First-order shear-deformation (Mindlin–Reissner) corrections would reduce the $n = 2$ frequency by approximately 4 % [31], a negligible effect compared with the material-property uncertainty. This limitation is unlikely to affect any conclusions.
5. **Equivalent-sphere geometry.** The oblate spheroidal abdomen is mapped to an equivalent sphere via $R_{\mathrm{eq}} = (a^2c)^{1/3}$. This preserves volume and leading-order curvature but suppresses the equatorial–polar mode splitting that occurs in an oblate spheroid [32]. For a typical aspect ratio $c/a = 0.67$, the splitting is estimated at ∼33 %, which could shift individual mode frequencies but would not change the coupling ratio or the qualitative separation between breathing and flexural mode families.

   More fundamentally, the human abdominal cavity is a *triaxial* ellipsoid (lateral ∼14 cm, craniocaudal ∼16 cm, anteroposterior ∼10 cm), not a spheroid of either kind. The oblate approximation captures the principal feature—anterior–posterior flattening—but any spheroid is a simplification. Crucially, however, the equivalent-sphere flexural frequencies depend on R_{eq} alone: at fixed R_{eq}, varying c/a from 0.3 (extremely oblate) to 3.0 (extremely prolate) changes f_2 by less than 1 %. The geometry enters the coupling ratio only through R_{eq} and the Ritz correction, so the results are robust to the choice of spheroidal idealisation.
6. **Inviscid fluid assumption.** The enclosed fluid is modelled as inviscid.

A viscous correction using the Stokes boundary-layer thickness shows that even at the worst-case mucus viscosity ($\mu = 0.1\,\mathrm{Pa\,s}$), the quality factor decreases by only $\Delta Q = -1.85\%$. The crossover viscosity at which viscous damping would equal structural damping is $281\,\mathrm{Pa\,s}$—the consistency of peanut butter. The inviscid assumption is therefore well justified for any physiologically realistic abdominal fluid.

7. **No internal organ structure.** Visceral organs are homogenised into a single fluid. An effective-medium analysis with solid inclusions (liver, spleen, kidneys) at a volume fraction $\phi = 0.25$ predicts a frequency shift of only $0.4\,\%$ for the $n = 2$ mode. The organ tissue bulk modulus ($K_{\mathrm{tissue}} \approx K_{\mathrm{water}}$) produces zero effective bulk modulus change, and at sub-percolation volume fractions the isolated inclusions cannot transmit shear. The homogeneous fluid model is therefore adequate for the present purposes.
8. **Gas pocket model simplifications.** The gas pocket analysis (§5.3) treats intestinal gas as spherical bubbles in an infinite homogeneous fluid, neglecting the cylindrical geometry of the gut lumen, viscoelastic wall loading, and the surrounding organ structure. These simplifications likely *overestimate* gas pocket displacement, because intestinal wall mass loading and stiffness would reduce the effective compliance. A coupled gas–tissue finite-element model is needed to refine these estimates.
9. **Mechanotransduction thresholds.** The $0.5\,\mu\mathrm{m}$ to $2.0\,\mu\mathrm{m}$ PIEZO activation range [18] derives from cell-culture experiments and may not directly translate to tissue-scale effects, where strain concentrations around cells are modulated by the extracellular matrix. This limitation introduces uncertainty in the absolute threshold but does not affect the *relative* comparison between airborne and mechanical coupling.

In summary, the identified limitations affect the absolute values of predicted frequencies and displacements—in several cases substantially—but none alter the central finding that airborne acoustic coupling to abdominal flexural modes

is 10^3–10^4 times weaker than mechanical coupling under occupationally relevant conditions.

5.8. Proposed Experimental Validation

We propose a two-stage experimental programme to validate the key predictions of the analytical model. The first stage employs a tissue-mimicking phantom: an oblate spheroidal silicone shell ($a = 0.18\,\mathrm{m}$, $c = 0.12\,\mathrm{m}$, $h = 10\,\mathrm{mm}$) filled with degassed water, matching the canonical geometry of §2. The shell material is platinum-cure silicone (Ecoflex 00-30 to Dragon Skin 10A), whose Young's modulus spans 0.07 MPa to 0.15 MPa—bracketing the relaxed abdominal wall estimate of $E = 0.1\,\mathrm{MPa}$. By casting phantoms at several mixing ratios, the $f_2 \propto \sqrt{E}$ dependence can be tested directly. Modal frequencies are measurable with an instrumented impact hammer or a low-frequency shaker sweep, and mode shapes can be reconstructed with a scanning laser vibrometer (Polytec PSV-500). The predicted $n = 2$ frequency for the recommended Ecoflex 00-30 formulation is 2.5 Hz; for Dragon Skin 10A, 3.6 Hz; the model predicts 4.0 Hz for the canonical human case.

The critical measurement is the coupling-ratio comparison. By mounting the phantom on an electrodynamic shaker (mechanical pathway) and separately exposing it to a subwoofer-generated infrasound field at 120 dB (airborne pathway), the displacement ratio $\xi_{\mathrm{mech}}/\xi_{\mathrm{air}}$ can be measured in a single experimental session. The model predicts ratios of order 10^3–10^4 across the silicone formulation range (Table 9). Non-detection of the airborne response at the vibrometer noise floor ($\sim 0.01\,\mu\mathrm{m}$) would itself constitute a positive validation of the weak-coupling prediction. The total cost of consumables (silicone, molds, sensors) is estimated at less than US$ 3 100, assuming access to shared laboratory equipment (shaker, laser vibrometer, DAQ).

The second stage extends to in vivo human subjects. Subject-specific abdominal wall moduli would be obtained via magnetic resonance elastography or shear-wave ultrasound elastography, and whole-body vibration transmissibility measured on an ISO 2631-compliant platform at sub-action-value amplitudes

($0.3\,\mathrm{m/s^2}$). A sample of 15 subjects, spanning a range of body habitus, is sufficient (power $= 0.80$, $\alpha = 0.05$) to detect a paired-test effect size $d = 1.0$ between measured and predicted peak-transmissibility frequencies. This programme requires Institutional Review Board approval and is estimated at 8–12 months, with the phantom study serving as a prerequisite validation of the model physics under controlled conditions.

6. Conclusions

The central result of this study is a coupling disparity of approximately 6.6×10^4 between mechanical whole-body vibration and airborne acoustic excitation of the human abdominal cavity. This single quantity, derived from first-principles shell theory and energy-consistent reciprocity analysis, resolves a long-standing empirical asymmetry: occupational whole-body vibration at 4 Hz to 8 Hz causes well-documented gastrointestinal effects, while airborne infrasound at comparable frequencies and extreme intensities does not.

The disparity has a clean physical origin. The abdominal cavity possesses two distinct mode families: a breathing mode ($n = 0$) near 2500 Hz, governed by the fluid bulk modulus and therefore irrelevant at infrasonic frequencies; and flexural modes ($n \geq 2$) in the 4 Hz to 10 Hz range that match ISO 2631 abdominal transmissibility data. Airborne sound at these frequencies has a wavelength exceeding 34 m, placing the abdomen deep in the Rayleigh regime ($ka \approx 0.01$). The $(ka)^n$ geometric penalty, compounded by the air–tissue impedance mismatch, reduces the effective driving pressure by four orders of magnitude. An energy-consistent analysis confirms an absorption efficiency of $\sim 10^{-14}$: the shell is, for practical purposes, acoustically transparent. Reaching the mechanotransduction threshold via airborne infrasound alone would require approximately 158 dB on the energy-consistent basis (135 dB using the less physical pressure-based estimate)—a level at which structural damage to the middle ear, blast lung injury, and building collapse would all present more immediate concerns than any intestinal indignity.

Mechanical vibration, transmitted through the skeleton and abdominal wall, couples directly and produces displacements 10^4 times larger at occupational exposure levels. This disparity is not a subtle biomechanical effect but a straightforward consequence of long-wavelength acoustics and impedance physics. Monte Carlo uncertainty quantification confirms the ratio spans 10^3–10^5 across physiologically realistic parameter ranges, with elastic modulus accounting for 86% of total variance.

The brown note, then, is a half-truth. The resonance it invokes is genuine; the mechanism it implies is not. What popular culture attributes to airborne sound is, in reality, the province of mechanical vibration—a distinction that matters considerably for occupational health standards and rather less for horror film screenwriters. Whether localised compressible inclusions (intestinal gas pockets) provide a secondary transduction pathway at extreme intensities remains an open question; they operate by quasi-static compliance, not by cavity resonance, and are explored separately.

Future work should validate the boundary condition assumptions through finite element analysis of anatomically realistic geometries, investigate the role of bowel gas pockets as localised acoustic transducers, and extend the model into the nonlinear regime relevant to high-amplitude mechanical excitation. In the meantime, concertgoers may rest assured—or, more precisely, remain seated—without fear of acoustic indignity.

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Data availability

All analytical models and computational scripts used in this study are available in the accompanying repository.

CRediT authorship contribution statement

Jonathan Mace: Conceptualization, Methodology, Supervision, Writing – Review & Editing. **Brian R. Mace**: Writing – Review & Editing, Validation. **GitHub Copilot CLI**: Software, Formal Analysis, Investigation, Data Curation, Writing – Original Draft, Visualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.